10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 8 — Biotechnology

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. Why are bacteria commonly used in genetic engineering for producing products?

(a) They naturally produce the required products
(b) They grow quickly and accept human genes
(c) They resist all infections
(d) They have a human-like immune system

Answer: (b) — Bacteria are simple, fast-growing and can easily accept new genetic material, so their cells can be given a human gene and made to produce the required product quickly. They do not naturally produce human proteins.

2. In genetic engineering, the first step is:

(a) Inserting DNA into the organism
(b) Cutting the desired gene from DNA
(c) Growing the modified organism
(d) Making proteins from the gene

Answer: (b) — First the gene of the donor organism is identified and isolated from its chromosome. Then the plasmid is cut and the gene is attached to it to make recombinant DNA, which is transferred into the bacterium.

3. Which of these is often used as a vector in genetic engineering?

(a) Virus
(b) Fungi
(c) Yeast
(d) Blood cell

Answer: (a) — The most commonly used vectors to transfer genes into bacteria are plasmids and bacteriophages, and bacteriophages are the viruses that can enter bacteria. Yeast and fungi are not used as vectors.

4. In genetic engineering, the desired gene is usually inserted into a bacterium by using:

(a) Protein
(b) Another bacterium
(c) Vector
(d) Enzyme

Answer: (c) — Genetic engineers use special vectors (plasmids or bacteriophages) to carry the desired gene into the bacterium. Enzymes are used only to cut and attach the DNA at specific locations.

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5. In genetic engineering, after inserting the gene, the host cell is allowed to:

(a) Mutate randomly (b) Die naturally

(c) Multiply and express the gene (d) Produce waste

Answer: (c) — After the recombinant DNA is put into the bacterium, the bacterium multiplies and makes a colony in which all the cells contain the recombinant DNA. These genetically modified bacteria then express the gene and produce the required protein.

6. In genetic engineering, “recombinant DNA” means:

(a) Natural DNA (b) DNA copied by mistake

(c) DNA joined from two sources (d) DNA destroyed by enzymes

Answer: (c) — The plasmid DNA is cut and the donor's gene is attached at its cut ends, so the new DNA is a combination of the original plasmid DNA and the new DNA. Such DNA, joined from two different sources, is called recombinant DNA.

7. A gene for toxin production is inserted into a plant for:

(a) Reducing water need (b) Increasing vitamin content

(c) Resisting insect attacks (d) Speeding up flowering

Answer: (c) — A gene from a bacterium is transferred into cotton and corn crops, and these genetically modified crops then produce a natural toxin which kills the insects but is safe for humans. Such crops are called Bt crops.

8. Genetically modified bacteria produce the following products, except:

(a) Hepatitis B vaccine (b) Growth hormone

(c) Insulin (d) Interferon

Answer: (a) — The Hepatitis B vaccine is produced from yeast cells through genetic modification, not from bacteria. Insulin, human growth hormone and interferon are produced in genetically modified bacteria.

9. Which biotechnology products help against viral infections?

(a) Antibiotics (b) Vaccines

(c) Clotting factors (d) Growth hormone

Answer: (b) — Vaccines are used to prevent viral and bacterial infections: they train the immune system to fight the real virus. Antibiotics are used against bacteria, not against viruses.

10. Golden rice is a genetically modified plant to increase:

(a) Protein (b) Vitamin A

(c) Iron (d) Calcium

Answer: (b) — A gene from a bacterium is transferred into rice plants, and the rice produces beta-carotene, which the body converts into vitamin A. Such genetically modified rice helps prevent blindness caused by vitamin A deficiency.

B. Write short answers.

1. What is the relation between biotechnology and genetic engineering?

Biotechnology is the use of living cells or organisms to create products or processes for specific purposes. Genetic engineering is the process of altering (editing) the genes of organisms, or transferring genes from one organism into other organisms. Genetic engineering is one of the main techniques of modern biotechnology; the other techniques include fermentation, tissue culture, cell culture and gene editing. So biotechnology is the wider field, and genetic engineering is an important tool used within it to make genetically modified organisms.

2. What is a plasmid? Why biologists use plasmids in genetic engineering?

A plasmid is a small, circular DNA molecule present in some bacteria. It is used as a vector, that is, as a carrier to introduce a new gene into a bacterial cell. Biologists use plasmids because they are small and easy to cut and join, they can carry a foreign gene into the bacterium, and they multiply along with the bacterium so that many copies of the gene are obtained. In this way bacteria can be made to produce useful proteins such as human insulin, growth hormone and vaccines.

3. The diagram shows how a gene is transferred to a bacterial cell. Identify the structures labelled 1 to 5.

Label	Structure
1	The desired gene (donor's gene) cut from the chromosome of the donor cell
2	The donor's gene after isolation from the donor cell
3	The plasmid removed from the bacterial cell and cut open
4	Recombinant DNA — the plasmid joined with the donor's gene
5	The genetically modified bacterium containing the recombinant DNA, which now multiplies and makes copies of the gene (gene cloning)

4. What do you mean by genetically modified organism?

A genetically modified organism (GMO) is an organism whose genes have been changed by genetic engineering, either by adding a gene from another organism, or by removing or replacing a part of its own DNA. As a result of genetic engineering the modified organism shows new characteristics, for example a Bt cotton plant which is resistant to insects, a rice plant which produces beta-carotene, or a bacterium which produces human insulin. So every organism produced by genetic engineering is called a genetically modified organism.

5. List two types of medical products that can be produced using genetic engineering.

(i) Human insulin, which is used for treating diabetes: the human insulin gene is transferred into a bacterium, and the modified bacteria produce insulin which is collected for the patients.

(ii) Vaccines, such as the vaccine against the Hepatitis B virus, which is produced from yeast cells through genetic modification, and protects people against this viral disease.

Human growth hormone (used to treat growth disorders) and interferon (an anti-virus protein) are other medical products made in genetically modified bacteria.

6. List two ways by which genetic engineering may improve crops.

(i) Resistance against insects and pests: a gene from a bacterium is transferred into cotton and corn crops, and the modified crops produce a natural toxin which kills the insect pests but is safe for humans.

(ii) Better nutrition: a gene is transferred into rice plants so that the rice produces beta-carotene, which the body converts into vitamin A; such golden rice helps to prevent blindness caused by vitamin A deficiency.

Genetic engineering can also produce crops that grow faster, resist diseases and herbicides, and survive drought better.

C. Write answers in detail.

1. Explain with examples that food biotechnology has advanced agriculture.

Biotechnology has improved agriculture in many ways, and the main examples are the following.

(i) Insect-resistant crops: a gene of a bacterium is transferred into cotton and corn plants. These genetically modified (Bt) crops produce a natural toxin that kills the insect pests but is safe for human beings, so the farmer needs fewer pesticides.

(ii) Herbicide-resistant crops: modified varieties of soybean, corn, tomato and cotton have been produced which show resistance against herbicides, so weeds can be removed easily without damaging the crop.

(iii) Disease- and drought-resistant crops: gene editing has produced crops which grow faster, resist pests and diseases and survive drought better; for example virus-resistant varieties of tomato and tobacco have been developed.

(iv) Better nutrition: golden rice has been produced by transferring a gene into rice, so the rice makes beta-carotene and helps to prevent vitamin A deficiency and the blindness caused by it.

(v) In Pakistan: research has led to the development of rice varieties which are resistant to certain diseases and pests, and tissue culture technology is used for the mass propagation of high-yielding fruit crops such as mangoes and citrus.

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2. Explain with examples that medical biotechnology has advanced healthcare in diabetes and cancer.

Diabetes: the human insulin gene is transferred into a bacterium with the help of a plasmid. The genetically modified bacteria now produce human insulin, which is collected and given to diabetic patients. In this way insulin is available in large quantity and without the risk of animal proteins, and the daily injections become safe and effective. In the Human Genome Project biologists also discovered the locations of all the genes in the human body and the characteristics controlled by each gene, which helps in identifying inherited diseases and genetic conditions early.

Cancer and other diseases: biotechnology is used to make proteins such as hormones and enzymes for treating diseases, to develop vaccines against viral and bacterial infections, to correct defective genes responsible for genetic disorders (gene therapy, for example for thalassaemia), and to produce targeted antibodies which are used for the diagnosis and treatment of diseases. Interferon, an anti-virus protein made in modified bacteria, is used to treat viral infections and boosts immunity, and gene editing is being researched for curing genetic diseases such as sickle cell anaemia and cystic fibrosis. More than 3,500 human disorders are caused by errors in genes, and biotechnologists are trying to remove such disorders through biotechnology.

3. Explain the role of yeast in the production of bread and ethanol.

Yeast is a microscopic fungus which carries out fermentation, and fermentation is the oldest form of biotechnology.

Role in bread making: yeast ferments the sugars present in the dough and produces carbon dioxide gas. This gas forms bubbles in the dough, because of which the dough rises and becomes soft and airy. The alcohol produced during this process evaporates during baking, so it does not remain in the bread.

Role in ethanol production: yeast ferments the sugars obtained from crops such as sugarcane or corn and produces ethanol (alcohol) and carbon dioxide. The ethanol is then purified and used as a biofuel, blended with petrol to produce a cleaner fuel, or used in alcoholic beverages. Yeast is also used to ferment plant starch into biodegradable plastic and to produce the Hepatitis B vaccine through genetic modification.

4. Enlist the steps for transferring a human gene into a bacterial cell.

(i) The gene of the donor organism (for example the human insulin gene) is identified and isolated from its chromosome.

(ii) A plasmid is removed from a bacterial cell and its DNA is cut. The donor's gene is attached at the cut ends of the plasmid, so the plasmid now contains its original DNA plus the new gene. This is called recombinant DNA.

(iii) The recombinant DNA is transferred into a bacterial cell with the help of the plasmid, which acts as the vector.

(iv) The bacterial cell is allowed to multiply and form a colony. All the genetically modified bacteria of the colony contain the recombinant DNA, so the colony contains many copies of the donor's gene; this step is called gene cloning.

(v) The modified bacteria express the gene and produce the required protein (for example insulin), which is then collected, purified and used for the patients.

5. Describe the advantages of gene editing.

(i) Curing genetic diseases: gene editing can correct the abnormal genes which cause diseases, and scientists are researching its use for sickle cell anaemia (a blood disorder) and cystic fibrosis (a lung disease).

(ii) Improving crops: gene editing can produce crops which grow faster, resist pests and diseases and survive drought better; virus-resistant varieties of tomato and tobacco have been developed, and modified soybean, corn, tomato and cotton resist herbicides.

(iii) Producing better medicines: gene editing helps to create bacteria which produce important medicines such as insulin for diabetes.

(iv) It is precise: it allows the scientists to add, delete or replace only a specific part of the DNA, so the required change is made without disturbing the other genes. In this way it can improve the health of human beings and the yield and quality of food crops.

6. Discuss the potential risks of genetic modification.

(i) Health concerns: some people worry that eating genetically modified foods could cause allergic reactions or long-term health problems, and that the new genes might interact with human cells in unexpected ways.

(ii) Environmental impact: GM plants can cross-pollinate with wild plants, so the modified genes may spread in natural ecosystems and create such species which may harm the environment.

(iii) Loss of biodiversity: if farmers mostly grow GM crops, then the traditional and naturally diverse varieties may become rare or extinct, and biodiversity is reduced.

(iv) Resistant pests and weeds: pests which are exposed again and again to GM crops may evolve into “powerful pests” which are no longer affected by the crop's built-in defences.

(v) Ethical issues: many people feel that it is morally wrong to alter the natural genes of living organisms. All these risks show that GM products must be tested carefully and used with proper rules.

D. Inquisitive questions

1. How can inserting a gene into a plant make it pest-resistant?

A gene which makes a toxin is taken from a bacterium and is inserted into the plant with the help of a vector. The plant cells now contain this gene, and when the plant grows it reads the gene and produces the natural toxin in its tissues. When an insect pest eats the leaves of the plant, this toxin enters the insect's body and kills it or stops it from growing and reproducing. The toxin does not affect human beings, because it acts only on the digestive system of the insect. In this way the plant protects itself from the pests without the need for chemical pesticides; such plants are called Bt crops.

2. How can biotechnology help in developing vaccines quickly, like for COVID-19?

Vaccines made by the older methods need a long time, because first the virus has to be grown and then weakened or killed. Modern biotechnology makes this much faster. The virus is first isolated and its genetic material is sequenced. Then only the gene which makes the surface protein of the virus is transferred into bacteria or yeast cells, which quickly multiply and produce that protein in large amounts for a protein vaccine. In another method, mRNA vaccines are made by using mRNA which carries the instructions to make harmless proteins of the virus, and this mRNA can be prepared in the laboratory very quickly. These proteins train the immune system to fight the real virus. Because microbes multiply very fast and the gene can be copied quickly, vaccines for a new disease such as COVID-19 can be developed in months instead of years.

3. Why is it important to carefully study the effects of genetically modified foods on health?

A genetically modified food contains a new gene and, therefore, a new protein which was not present in that food before. This new protein may cause an allergic reaction in some people, and its long-term effects on the human body are not fully known. Sometimes the new gene may also change the amount of natural substances in the food. Therefore, before such a food is allowed in the market, its safety must be tested thoroughly on animals and then studied in human beings, so that any harmful effect on health is known in advance. Careful study also protects the environment, because GM plants may cross-pollinate with wild plants and spread their genes. This is why GM foods are labelled and regulated by government authorities.

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